TABLE I

NEGATIVE RESULT ON A LEARNED RE-RANKER BASELINE (CAR, A SINGLE-HEAD CROSS-ATTENTION RE-RANKER OVER THE TOP-K CANDIDATES) ON THE ORIGINAL 6-SETUP CROSS-CONDITION GRID (OXFORD-NIGHT AND NCLT-2012-10-28 $\times$ 3 BACKBONES). FIVE STANDARD FIXES ARE LISTED IN §. ALL LEARNED VARIANTS FAIL TO CONTROL FDR; SAFEVPR SUCCEEDS ON THE SAME SETUPS.

	$\alpha = 0.05$			$\alpha = 0.10$		
Method	Pass	FDR	TPR	Pass	FDR	TPR
CAR (cross-attention re-ranker, 118 k params)	1/6	0.1366	0.736	1/6	0.2384	0.847
CAR + within-query z-score	1/6	0.1447	0.705	1/6	0.2562	0.862
CAR + test-time temperature scaling	1/6	—	—	1/6	—	—
CAR + DANN (gradient reversal)	0/6	—	—	0/6	—	—
CAR + IWCP	1/6	—	—	1/6	—	—
SAFEVPR (ours)	6/6	0.0039	0.503	6/6	0.0152	0.616